

CENTRAL WORKSHOP SAFETY MANUAL

(For Students and Trainees)

GENERAL SAFETY RULES (Applicable to All Sections)

- Always Wear proper Personal Protective Equipment (PPE): safety shoes, goggles, gloves, and lab coats.
 - Do not operate any machine without authorization or supervision.
 - Maintain cleanliness and order in the workspace.
 - Report any unsafe condition or accident immediately to the instructor.
 - Avoid loose clothing, jewellery, and untied hair.
 - Do not engage in horseplay or distractions.
 - Know the location of first aid kits, fire extinguishers, and emergency exits.
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1. FITTING SHOP SAFETY

Hazards:

- Sharp tools and metal edges
- Flying chips during filing or cutting

Safety Guidelines:

- Use tools like files, hacksaws, and chisels properly.
 - Always use goggles while chipping or filing.
 - Ensure proper clamping of workpieces in vices.
 - Do not use damaged tools.
 - Keep hands away from cutting edges.
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2. FABRICATION SHOP SAFETY

Hazards:

- Sharp metal sheets
- Heavy materials and tools

Safety Guidelines:

- Wear cut-resistant gloves when handling sheets.

- Use proper lifting techniques for heavy materials.
 - Keep working area free of scrap and clutter.
 - Use correct tools for bending, cutting, and shaping.
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3. WELDING SHOP SAFETY

Hazards:

- Intense heat and sparks
- Harmful radiation and fumes

Safety Guidelines:

- Always wear a welding helmet with proper filter glass.
 - Use flame-resistant clothing.
 - Ensure proper ventilation or exhaust systems.
 - Keep flammable materials away.
 - Do not touch hot metal surfaces.
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4. MACHINE SHOP SAFETY

Hazards:

- Rotating machinery
- Flying chips and entanglement

Safety Guidelines:

- Do not wear loose clothing near machines.
 - Ensure guards are in place before operating machines.
 - Never leave machines running unattended.
 - Use brushes, not hands, to remove chips.
 - Turn off machines before making adjustments.
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5. CNC SECTION SAFETY

Hazards:

- Automated machine movements
- High-speed operations

Safety Guidelines:

- Only trained students may operate CNC machines.
 - Check program code before execution.
 - Keep machine doors closed during operation.
 - Do not attempt to stop the machine manually.
 - Use emergency stop when required.
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6. 💧 HYDRAULIC & PNEUMATICS SAFETY

Hazards:

- High-pressure fluids and air
- Sudden release of pressure

Safety Guidelines:

- Check hoses and fittings for leaks before operation.
 - Never exceed recommended pressure limits.
 - Release pressure before maintenance.
 - Keep hands away from moving actuators.
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7. ⚡ ELECTRICAL SECTION SAFETY

Hazards:

- Electric shock
- Short circuits and fire

Safety Guidelines:

- Ensure hands are dry before handling equipment.
 - Do not touch live wires.
 - Use insulated tools.
 - Switch off power before making connections.
 - Report damaged wires immediately.
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8. ELECTRONICS SECTION SAFETY

Hazards:

- Sensitive components
- Static electricity and minor shocks

Safety Guidelines:

- Handle components carefully to avoid damage.
- Use anti-static wrist straps where necessary.
- Avoid short circuits on circuit boards.
- Ensure proper polarity before connections.
- Keep liquids away from electronic devices.

EMERGENCY PROCEDURES

- In case of injury: Inform instructor and use first aid kit.
- In case of fire: Use appropriate fire extinguisher.
- In case of electric shock: Switch off power immediately.
- Evacuate calmly during emergencies.

CONCLUSION

Safety is everyone's responsibility. Following these guidelines ensures a safe and productive workshop environment for all students.
